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1. (**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file here. If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

1 / 1 point

Let

$$f(\alpha, \beta) = \sum_{i=1}^{n} \left[y_i - (\alpha + \beta^{\mathrm{T}} x^i) \right]^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

be the objective function of ridge regression. Which of the following is $\frac{\partial f(\alpha,\beta)}{\partial \beta_j}$?

- ( ) $2\sum_{i=1}^{n}[y_i - (\alpha + \beta^{\mathrm{T}} x^i)]x_j^i + 2\lambda$.
- ( ) $-2\sum_{i=1}^{n}[y_i - (\alpha + \beta^{\mathrm{T}} x^i)]x_j^i + 2\lambda$.
- ( ) $2\sum_{i=1}^{n}[y_i - (\alpha + \beta^{\mathrm{T}} x^i)]x_j^i + 2\lambda\beta_j$.
- (•) $-2\sum_{i=1}^{n}[y_i - (\alpha + \beta^{\mathrm{T}} x^i)]x_j^i + 2\lambda\beta_j$.
- ( ) None of the above.

Correct

2. What is the outcome of projecting the vector $(1, 1)$ onto the vector $(2, 1)$?

1 / 1 point

- ( ) $(2, 1)$.
- ( ) $(\frac{6}{\sqrt{5}}, \frac{3}{\sqrt{5}})$.
- (•) $(\frac{6}{5}, \frac{3}{5})$.
- ( ) $(\frac{4}{5}, \frac{2}{5})$.
- ( ) None of the above.

Correct

3. Consider three data points $(2, 1)$, $(4, 2)$, and $(1, 3)$, where the first two belong to class 1 (say, success) and the last one belongs to class 2 (say, failure). The SVM problem to classify these three points can be formulated as

1 / 1 point

$$\begin{array}{rl} \min\limits_{\alpha, \beta_1, \beta_2} & \frac{1}{2}\left(\beta_1^2 + \beta_2^2\right) \\ \text{s.t.} & \alpha + 4\beta_1 + 2\beta_2 \geq 1 \\ & \alpha + 2\beta_1 + \beta_2 \geq 1 \\ & \alpha + \beta_1 + 3\beta_2 \leq -1. \end{array}$$

Which of the following is an optimal solution to the above nonlinear program?

- ( ) $(\alpha, \beta_1, \beta_2) = (1, 0.4, 0.8)$.
- (•) $(\alpha, \beta_1, \beta_2) = (1, 0.4, -0.8)$.
- ( ) $(\alpha, \beta_1, \beta_2) = (1, -0.4, 0.8)$.
- ( ) $(\alpha, \beta_1, \beta_2) = (-1, 0.4, -0.8)$.
- ( ) None of the above.

Correct

4. Continue from the previous question. For such an instance, SVM gives us a line to separate the two classes perfectly, and that line may be depicted on a two-dimensional figure with the three data points. What is the intercept of this line?

1 / 1 point

- ( ) 1.
- (•) $\frac{5}{4}$.
- ( ) $\frac{3}{2}$.
- ( ) -1.
- ( ) None of the above.

Correct

5. For the following statements, select all that are correct:

1 / 1 point

- [x] The primal SVM problem disallowing violation is a convex program.

  Correct

- [x] The primal SVM problem disallowing violation may be infeasible.

  Correct

- [x] The primal SVM problem allowing violation is a convex program.

  Correct

- [ ] The primal SVM problem allowing violation may be infeasible.
- [x] The dual SVM problem is a convex program.

  Correct